ABSTRACT

ChaiNGO:

The aim of this project is to increase accountability and trust in the way NGOs Receive and Organize their funds. We ensure this by safely documenting every transaction on an unchangeable digital ledger by utilizing blockchain technology, these records can then be retrieved later whenever needed to verify the authenticity of the transactions.

Every transaction is assigned a unique ID, and the user-friendly interface makes it easy for Patrons and the NGOs to view each transaction. Administrators can handle funds and produce reports and donors can see how the funds that they have donated are being utilized.

The platform has three primary modules: **Admin Module:** It assists the NGO administrators in managing finances, monitoring spending, and supervising activities.
Donor Module: It enables the donors to make transparent contributions, view the utilization of their funds, and receive project updates.
NGO Module: It enables the NGOs to register their organisation, projects and collect donations. They can also use it to send out updates, show fund utilization, etc.

This project contributes to the development of trust between NGOs and their supporters by ensuring that money transactions are safe and transparent. In addition to encouraging appropriate financial procedures in the NGO sector, it guarantees that donations are used as intended.

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Problem Statement

S.NO	Solution Models	Limitations	Advantages
[1]	Block Chain, AI model	most NGOs remain reluctant to share information, and many refuse to recognize the need for accountability	Demonstrates transparent use of resources to donors and regulators.
[2]	Block Chain	NGOs Struggle due to a lack of Transparency In Their actions	Demonstrates Transparent Actions
[3]	Block Chain	Securing consistent and reliable funding is a pervasive challenge for NGOs.	Demonstrates a Reliable Funding Option For NGOs
[4]	Block Chain	Operating in conflict zones or areas with political instability exposes NGO staff to significant security threats	Demonstrate a safe and reliable option For NGO operations

- **Lack of Transparency**: The Donor's don't have a clear idea on how the donated funds are being used which could cause mistrust and other issues among donors and NGOs.

- **Risk of Fraud**: Normal Or Non Block-chain based organizations have a high risk of allowing various nefarious activities with the funds being donated like embezzlement or fraud.

- **Lack of Fund Tracking**: NGOs struggle to show how the funds being donated are being utilized towards their goal,which in turn could create trust issues between the Donor's and the organizations.

- **High administrative cost**: Managing transactions manually can be very time consuming and expensive for the organizations to bear.

- **Lack of Accountability**: There is a lack of accountability for how donations are allocated and spent by the organization, which can cause a lack of trust between the Donors and the NGOs

- **Difficulty Auditing**: There could arise a difficulty in auditing due to the work required to do manual auditing of the records held by the organization.

- **Donor engagement:** Donor's could be left clueless about how their donation is being put into effect due to the lack of engagement by the Organizations by providing regular updates of the work being done.

By addressing these pressing issues, the project "Blockchain-Based Transaction Transparency for NGOs" aims to build trust, improve accountability, and ensure efficient utilization of resources through a secure, decentralized system.

CHAPTER 1
INTRODUCTION

1.1 PURPOSE
1.2 SCOPE
1.3 DEFINITIONS, ACRONYMS, and ABBREVIATIONS
1.4 REFERENCES
1.5 OVERVIEW

1.1 PURPOSE

A. **Traditional System and its limitations:**

   NGOs rely on centralized databases and human data entry to maintain their donation records. These entries include:

   a. **Donation Records**: Donations received through bank transfers or payment gateways like Razorpay, PayPal, etc. These records are stored in private databases
   b. **Expenditure Records**: NGOs manually prepare reports of their expenditure using spreadsheets or other tools to show how the donations that they have received are being spent.
   c. **Auditing Records**: Audits and annual reports by external consultants are prepared to build credibility and increase trust in the organisation. These reports are often prone to human error and tampering.

   Limitations of Traditional System:

   a. **Lack of Transparency**: Donors cannot ensure if the funds that they donated are being used for their right purpose.
   b. **Risk of Fraud**: Records that are prepared by humans can be tampered with or falsified.
   c. **Inefficient use of resources**: Preparing reports and maintaining records manually can take a lot of time and resources which could be utilized elsewhere.
   d. **Increasing Costs**: Hiring external auditors increases the cost of operations for the NGOs, this money could rather be used for welfare activities.
   e. **Lack of Accountability**: NGOs need to show the impact that they have created through their activities which is quite difficult to demonstrate.

B. **Blockchain Integration**

   To tackle the above mentioned issues, we propose to integrate blockchain technology into our platform in the following ways:

   a. **Donation Tracking**: Transactions made on the blockchain cannot be changed, therefore it is a credible way to record all the transactions.
   b. **Implementing Smart Contracts**: Automation of release of funds to NGOs based on the completion of their projects or certain milestones which can be verified by a third party.
   c. **Decentralized Storage**: All the project updates, receipts, etc can be stored on the blockchain or using decentralized storage such as IPFS.
   d. **Tokenizing Transactions**: All the donations can be represented using ERC-20 Tokens for added traceability.

C. **Solutions using Blockchain**

a. **Fund Utilization** -> Transparent Ledgers: Donors can track their funds in real time using blockchain explorers.
b. **Manual Delays** -> Automated updates: NGOs can submit progress reports verified by third party auditors on-chain.
c. **Fraudulent Record Keeping** -> Data Immutability: Data on the blockchain cannot be tampered with or changed.
d. **High Costs** -> Eliminating Manual Auditors: Smart Contracts can replace manual audits and payment gateways/banks which will drastically reduce operation costs.
e. **Donor trust** -> Proof of impact: Donors receive an on-chain proof of the funds donated which can mitigate trust issues.

D. Proposed Working Model

The platform will operate as listed below:

a. **Donor Registration**: Donors create a crypto-wallet (eg.MetaMask) and undergo KYC and AML checks to verify the authenticity of the donor
b. **Project registration**: NGOs can register projects, set milestones and link them to smart contracts.
c. **Donation Management**: Donors contribute via Crypto/fiat and funds are locked in escrow or in other words held by a middleman until certain conditions are met.
d. **Donor Portal**: Donors can access a dashboard to view transactions, project updates and impact metrics.

E. Models and Modules Used:

Modules:

a. **Donor Module**: Crypto-wallet integration, donation tracking and impact statistics.
b. **NGO Module**: Project registrations, goals management and statistical tools.
c. **Admin Module**: KYC/AML Compliances, login management and all such backend tasks.
d. **Blockchain Module**: Transaction management, smart contract management (Goals, escrow, etc) and decentralised storage (IPFS, etc)
e. **Reporting Modules**: Generation of transparency report and analytics

Technical Models:

a. **ERC-20 Tokens**: To represent donations, transactions and rewards.
b. **IPFS**: Decentralised storage for files such as reports, audits, project photos, etc

1.2 SCOPE

This project will be used by various parties such as NGOs and other Non Profit Organisations to manage and ensure transparency and accountability in transactions and allocation of funding provided by various donors for charity in a transparent interface.
The lack of transparency is one of the major reasons for the donors to be unsure about the usage of the funds that they have provided. This platform aims to create a safe and transparent space free from hidden activities that ensures the patrons that their money is being used for the right purpose.

1.3 DEFINITIONS

a. NGO: Non Governmental Organisation.
b. Web3: Decentralised internet based on blockchain.
c. dApp: Decentralised Application (Web3 Application)
d. Smart Contract: Self Executing program that is set to trigger upon meeting the requirements as specified by its code.
e. Cryptocurrency: Digital currency based on blockchain technology.
f. Donor: An individual/organisation that provides donations to the NGOs.
g. SRS: Software Requirement Specification
h. DFD: Data Flow Diagram
i. UI/UX: User Interface / User Experience
j. API: Application Programming Interface
k. KYC: Know Your Customer (verification process as mandated by guidelines)
l. AML: Anti Money Laundering (regulatory compliance as mandated by guidelines)

1.4 REFERENCES

1. Science Direct Research on NGOs
2. Importance of transparency in NGO Activities
3. Overcoming NGO Failures and building resilience
4. Harsh conditions for Aid Workers

1.5 OVERVIEW

Our application consists of three main modules: the Donor Module, the NGO Module, and the Admin Module.

A. **Donor Module:**
    a. Donors can create accounts, browse various projects listed by the NGOs and make donations to the project(s) of their choosing.
    b. They can track their donation in real-time, view reports that are automatically generated and check the proof of impact of the donation that they have made.
    c. Donors can also interact with the NGOs through the platform, providing feedback and also to request additional information.

B. **NGO Module:**
    a. NGOs can register their organisation on the platform, list projects and receive donations for the users.
    b. NGOs can provide updates on project progress, utilization of funds and the impact statistics.
    c. NGOs can use smart contracts to automate release of their funds based on reaching a certain goal therefore ensuring accountability.

C. **Admin Module**:
    a. Admins oversee the platform, verify NGO registration and ensure their compliance with KYC and AML regulations.
    b. They can generate reports, monitor transactions and resolve any platform issues.
    c. Admins can also provide platform updates and manage users.

Advantages:

a. **Transparency**: All transactions and activities are recorded on the blockchain, ensuring accountability.
b. **Trust**: Donors can verify how their funds are used, increasing trust in NGOs.
c. **Efficiency**: Automation through smart contracts reduces administrative overhead.
d. **Security**: Blockchain ensures data integrity and protection against tampering.

e. **Global Reach**: Cryptocurrency donations enable cross-border contributions without intermediaries.

Disadvantages:

a. **Complexity**: Blockchain technology may be difficult for non-technical users to understand.
b. **Scalability**: Handling a large number of transactions on the blockchain can be challenging.
c. **Regulatory Challenges**: Compliance with local laws and regulations for cryptocurrency usage.
d. **Adoption Problems**: NGOs and donors may need education and training to use the platform effectively.

CHAPTER 2
SOFTWARE REQUIREMENT SPECIFICATION

2.1 Product Perspective

This Blockchain based NGO Transaction project is a platform system that manages and tracks financial activities as a clear ledger with a well structured and definite perspective on who and where the funds of individuals are going to.

Due to the improperly managed transactions when it comes to NGO's the users face quite a lot of difficulties in accessing past data as well as managing present data.
A fully transparent and traceable NGO transaction tracking system which will be developed through this project will eliminate the disadvantages caused by the legacy system by improving the reliability,ease of payment and tracking , flexibility and efficiency. The usage of Blockchain to digitize and put all the transactions into a ledger will accommodate the needs of the user effectively.

2.1.1 System Interfaces

User Interface

A) **Admin Interface:** Backend system for platform administrators to verify NGO details, monitor transactions and manage accounts. The interface includes:
    a. User and NGO verification
    b. Transaction Monitoring Panel
    c. Database Monitoring

B) **NGO Interface:** A platform for NGOs to manage tier organisation profile, create and update projects, receive donations and give updates to their donors. The interface features will include:
    a. Profile Management
    b. Project Creation Tools
    c. Fund Tracking
    d. Donor communications and milestone tracking

C) **Donor Interface**: A responsive Web-App with a dashboard for donors to register, browse projects, make donations and track their contributions. The interface will include:
    a. Navigation Menu
    b. Donation Forms
    c. Transaction History
    d. Profile Management

Hardware Interfaces

A) **Hardware Crypto-Wallet Integration:** Interface for integration of hardware wallets like Ledger and Trezor through standard libraries for secure transactions.

Software Integration

A) **Blockchain Network:** Interface with Sepolia test-chain network for transaction processing and smart contract deployment.
B) **Social Media APIs:** Integration to share projects for wider reach and engagement.
C) **Emails Service Providers:** To provide emails notifications and updates.

2.1.2 System Specifications

2.1.2.1 H/W Requirements

- ➔ Core i3 Processor.
- ➔ 4GB RAM
- ➔ 1TB of storage space in the ser
- ➔ +-ver machine.

2.1.2.2 S/W Requirements

- ➔ Database: MongoDB 5.0 or higher
- ➔ Windows 7 or higher
- ➔ Blockchain Network: Sepolia Test Net
- ➔ Smart Contract Development: Solidity 0.8.0+
- ➔ Backend: Node.js and Express.js
- ➔ IDE: Visual Studio Code.
- ➔ Deployment: ThirdWeb and Local website deployment

2.1.3 Communication Interfaces

- ➔ RESTful APIs: For frontend and backend interaction.
- ➔ Web3.js: For blockchain integration.
- ➔ Notification APIs: For emails and updates.
- ➔ Social Media APIs: For social media integration.
- ➔ Blockchain Explorer APIs: For transaction verification, monitoring and viewing.

2.2 Product Function

Donor Module Functions

1. User Registration and Login
2. Project Catalogue Browsing
3. Donation Management
4. Milestones and Updates tab

NGO Module Functions

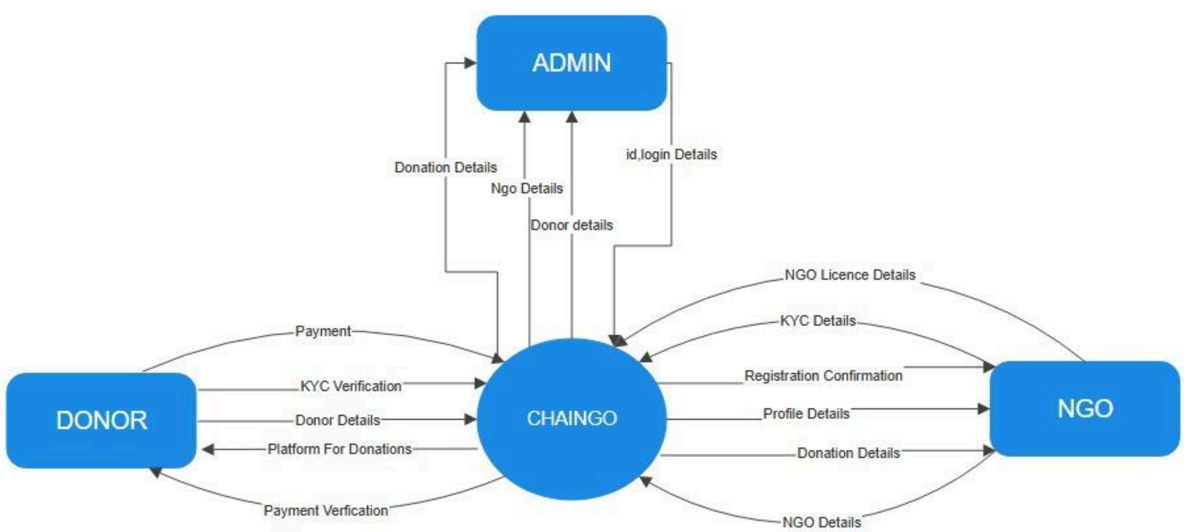
1. Organisation Registration
2. Project Management
3. Fund Management and Tracking
4. Donor updates

Admin Module Functions

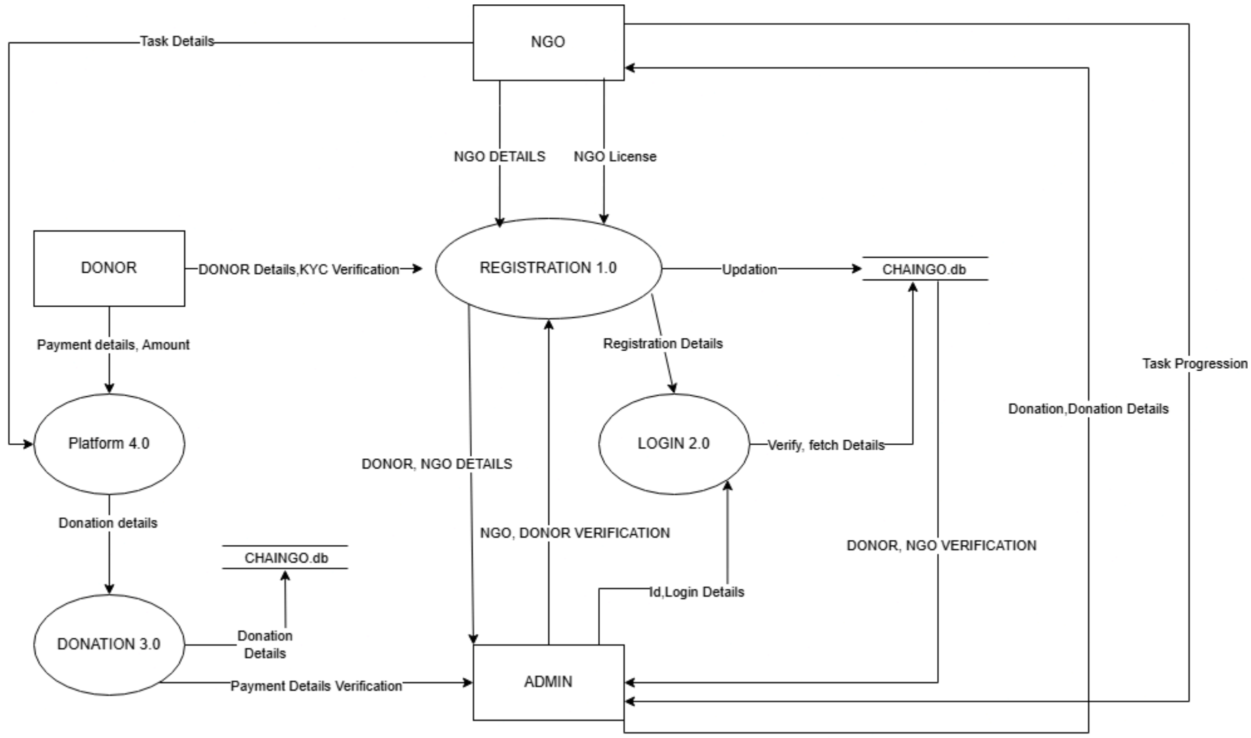
1. Platform Management
2. Verification and User Management
3. Fund Monitoring
4. Platform Administration

2.3 Data Flow Diagram (DFD)

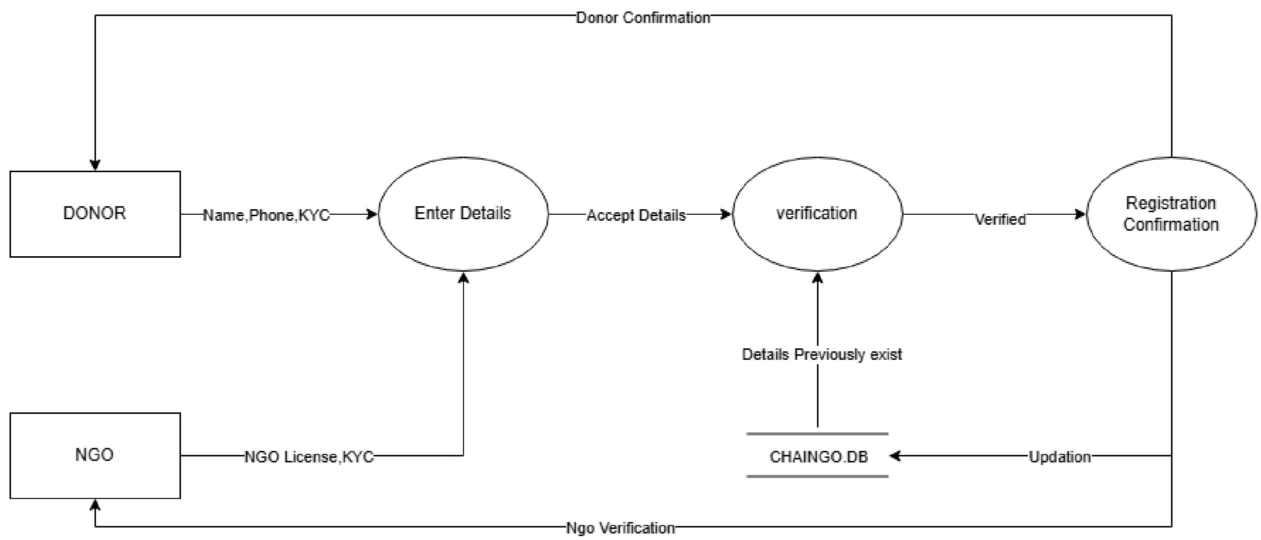
Context Level Diagram:



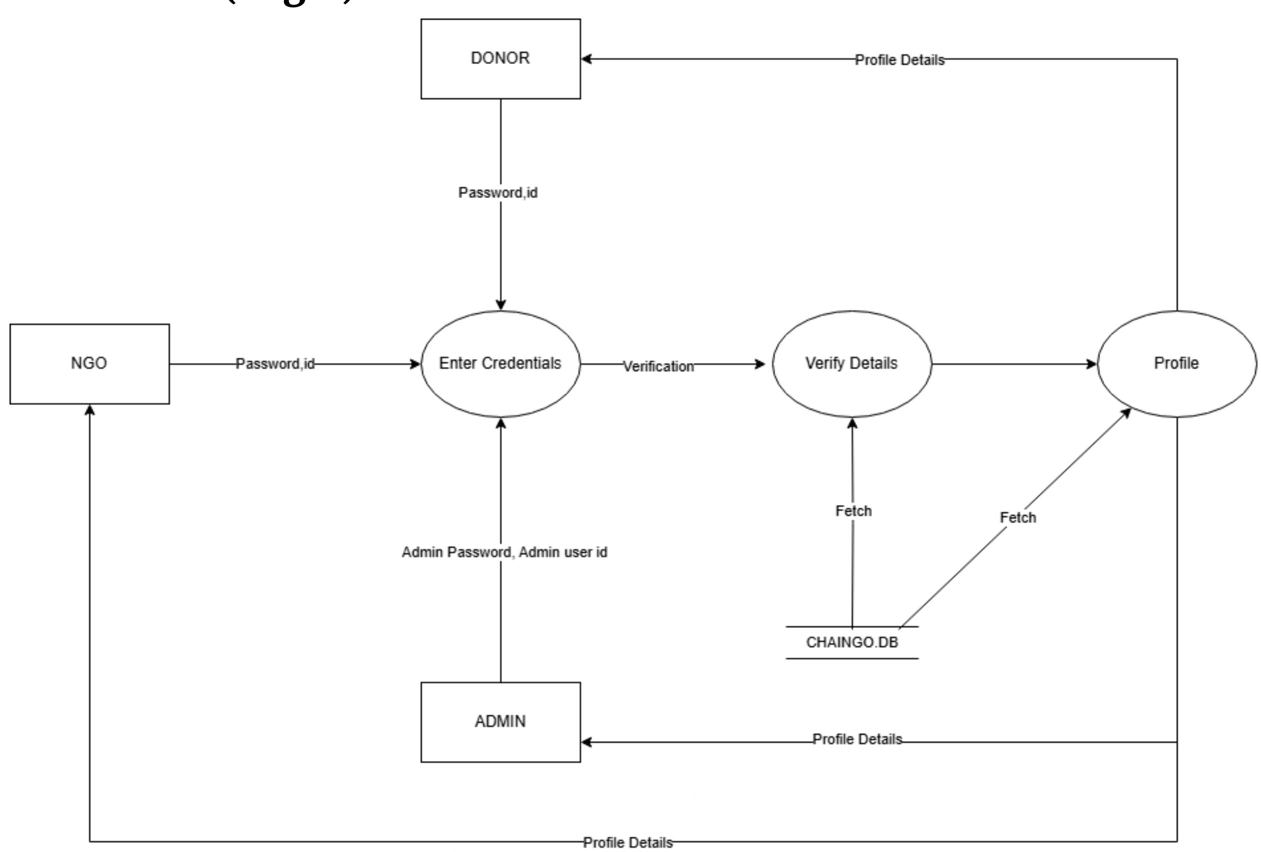
DFD Level 1:



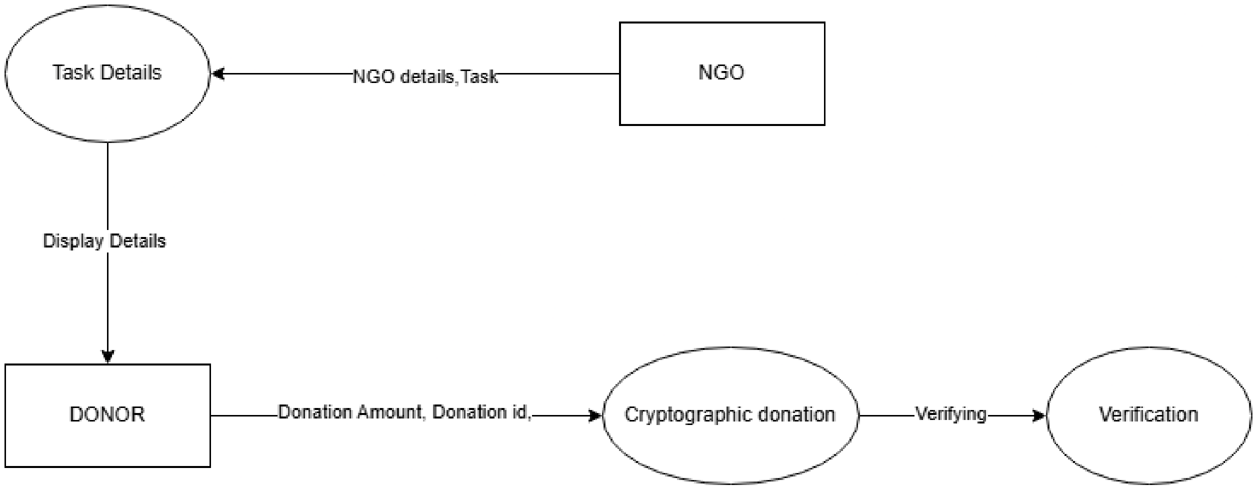
DFD Level 2(Registration):



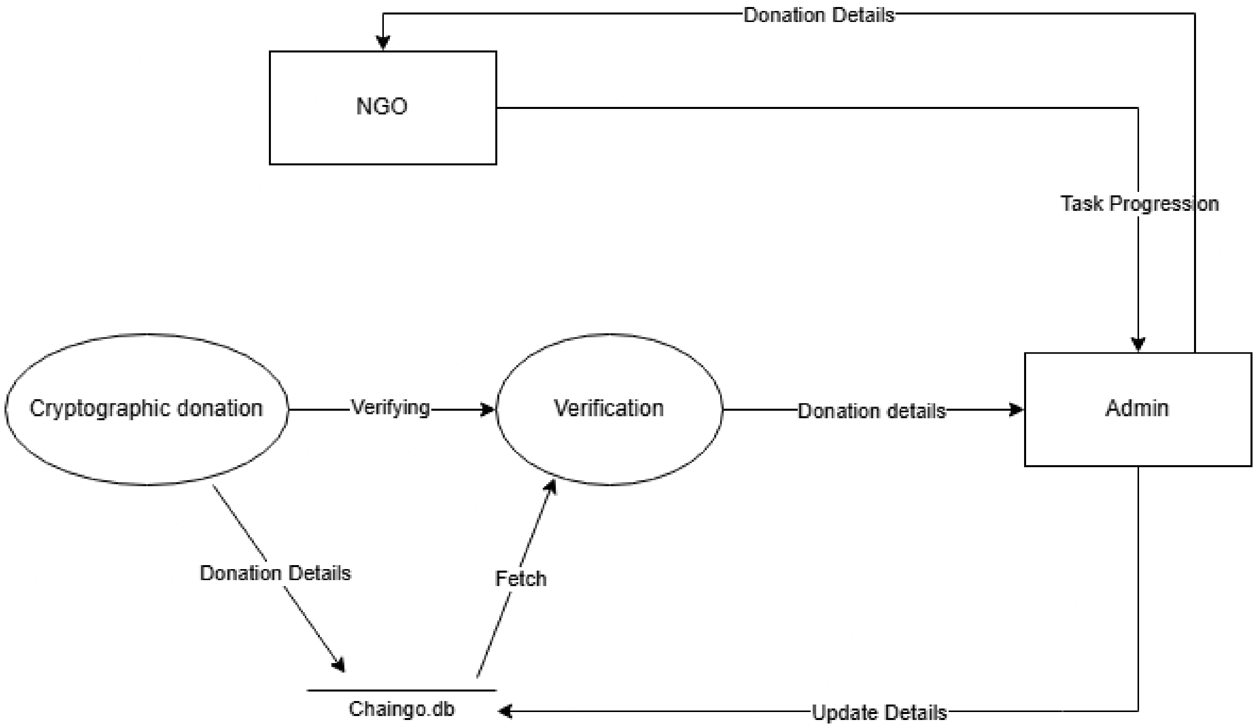
DFD Level 3(Login):



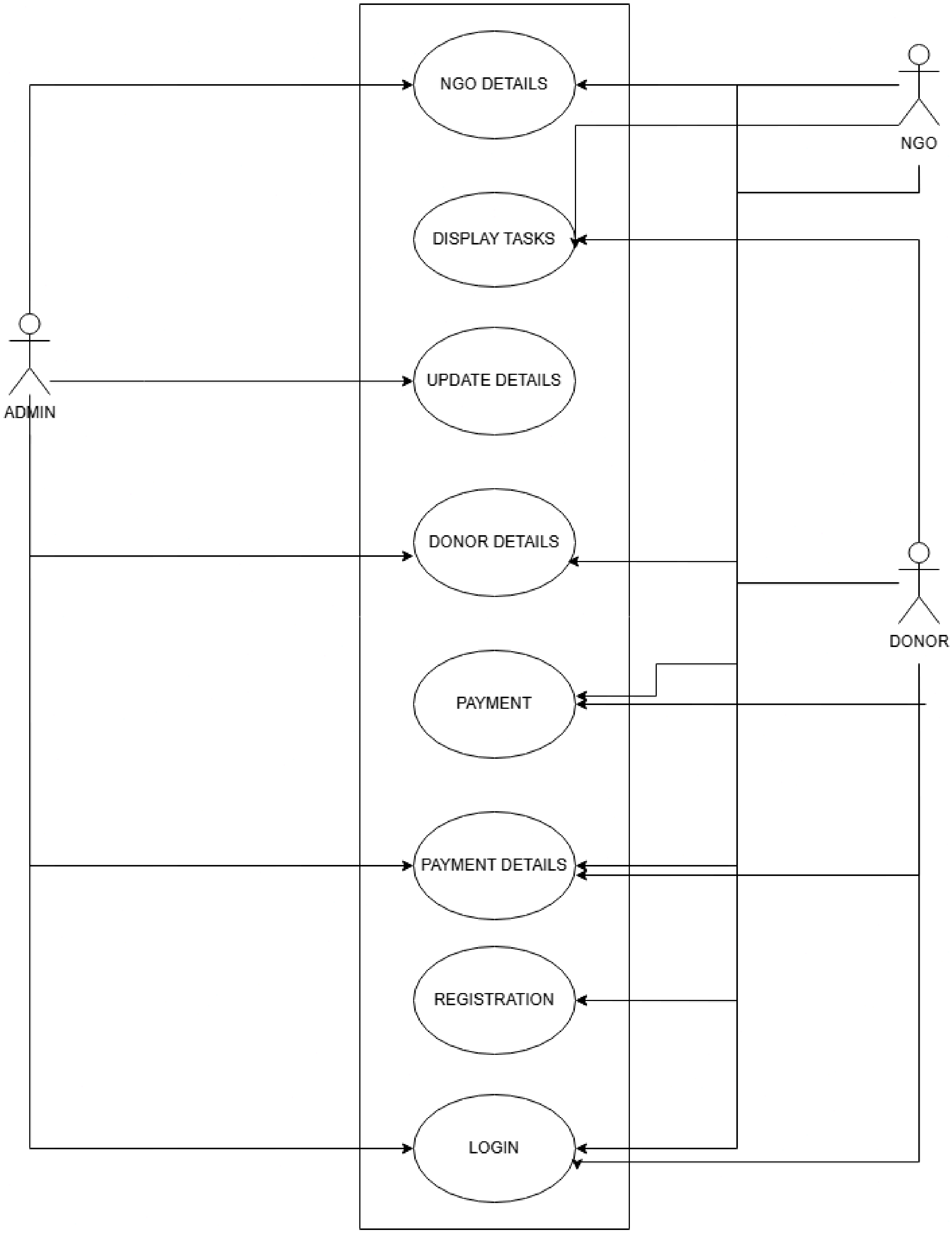
DFD Level 4(Platform):



DFD Level 5 (Donation):



2.4 Use Case Diagram


NGO DETAILS
DISPLAY TASKS
UPDATE DETAILS
DONOR DETAILS
PAYMENT
PAYMENT DETAILS
REGISTRATION
LOGIN
ADMIN
NGO
DONOR

.5 Use Case Description

(1) Donor

Description: Allows a new donor to create an account on the platform

Pre-conditions: User has internet access and valid email

Main Flow:

1. User accesses the registration page
2. User provides personal information and email
3. User completes KYC verification
4. User connects or creates crypto wallet
5. System verifies information and creates account
6. User receives confirmation email Post-conditions: User has access to donor features

Alternative Flows:

- Email already registered: System notifies user and suggests login
- KYC verification fails: System provides reasons and retry options

(2) NGO Representative

Description: Allows NGOs to register on the platform

Pre-conditions: NGO has required documentation

Main Flow:

1. NGO representative accesses NGO registration page
2. Representative provides organizational details
3. Representative uploads required legal documents
4. Representative provides banking and wallet information
5. System stores information for admin verification
6. Admin reviews and approves NGO

Post-conditions: NGO has access to organizational features

Alternative Flows:

- Incomplete documentation: System requests missing documents
- Verification fails: System notifies with reasons

(3) Donor

Description: Allows donor to contribute funds to projects

Pre-conditions: Donor has registered account with verified status

Main Flow:

1. Donor browses or searches for projects
2. Donor selects project to support
3. Donor specifies donation amount and payment method
4. System processes transaction on blockchain
5. Smart contract locks funds in escrow
6. System generates receipt and transaction ID

Post-conditions: Donation is recorded on blockchain

Alternative Flows:

- Payment failure: System provides alternative payment options
- Insufficient funds: System notifies donor

(4)\NGO

Description: Allows NGO to create new funding projects

Pre-conditions: NGO has verified account

Main Flow:

1. NGO accesses project creation page
2. NGO provides project details (title, description, goals)
3. NGO sets funding targets and milestones
4. NGO uploads supporting media and documents to platform
5. NGO configures smart contract conditions
6. System publishes project after review

Post-conditions: Project is available for donors to view and fund

Alternative Flows:

- Incomplete information: System requests missing details
- Project rejected: Admin provides feedback for revision

(5) Donor

Description: Allows donor to monitor how their funds are used

Pre-conditions: Donor has made at least one donation

Main Flow:

1. Donor accesses their dashboard
2. Donor views list of supported projects
3. Donor selects specific project or donation
4. System displays real-time fund utilization data
5. System shows project milestones and progress
6. Donor can view proof of impact documentation

Post-conditions: Donor has current information on fund usage

Alternative Flows:

- No updates available: System shows last known status and update timeline

2.6 User Characteristics

ADMIN

The admin has full access to the transaction logs and Ensures safe and smooth transactions. It also manages and verifies NGOs and users. They are the highest privileged user who can access the system.

Key functions:

- Verifies NGOs and users
- Monitors transactions
- Manages the database

DONOR

The donor can donate to the NGOs and can view all current transactions.They get provided with an easy way to browse from a catalogue of available NGOs and can track their donation history.

Key functions:

- Browse NGO projects
- Make donations easily
- Track donation history
- Register and manage Profiles

NGO

After the NGO is registered it can manage projects and donations. The NGO is given tools to create and update projects. They can also communicate with the users with a clear display of transactions to build trust.

Key Functions:

- Creates and updates profile
- Track received donations
- Send updates and Milestones to users
- Can manage the organization's profile

2.7 Constraints

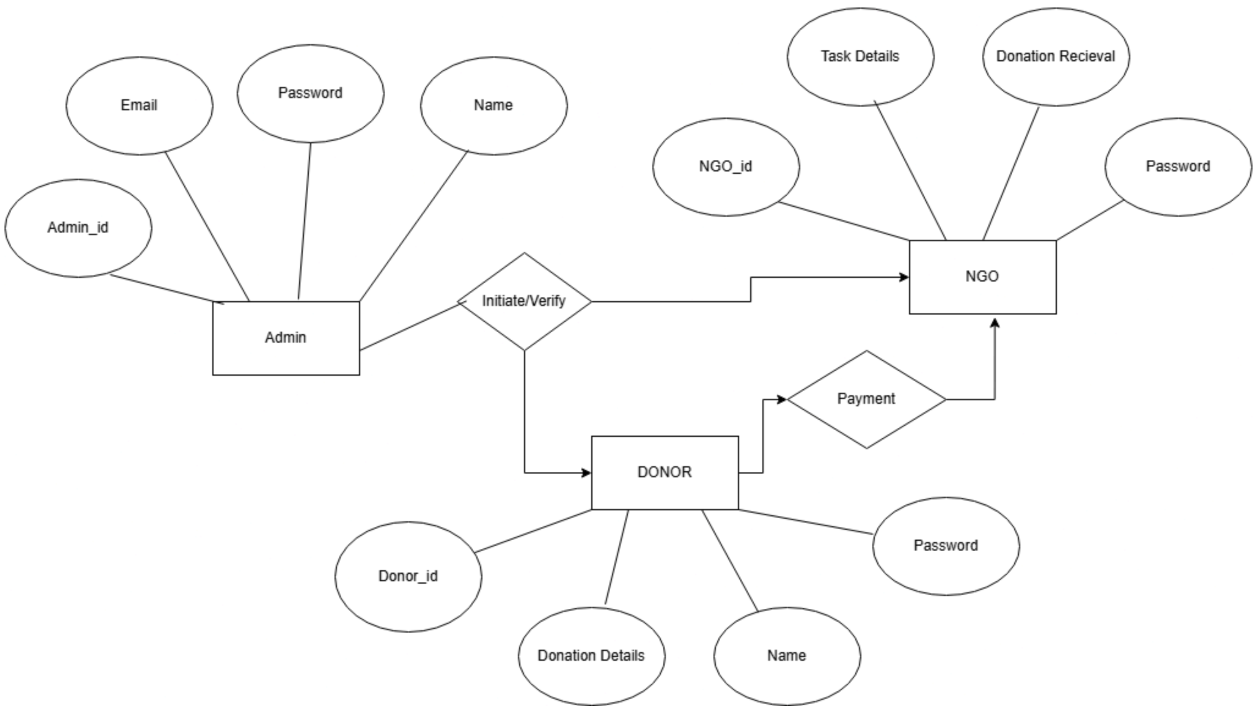
- Cross platform support for various devices and browsers
- Managing blockchain transactions fees and challenges faced when migrating to new blockchain technologies.
- Handling high traffic during donation rallies or campaigns.
- Regular Backup with minimum Downtime.
- Must work with pre-existing NGOs financial System and should sync with pre-existing data without issues .

2.8 Assumptions & Dependencies

- Only the Admin has permission to delete records
- NGOs can update the project details but they can't delete user donation records.
- Users can view their own transaction history but they can't edit or delete it.

- Each user must have a valid User ID and a Password and they should log in to access any records or perform any tasks.

ER Diagram



ER TABLE

Admin

SNO	Column Name	Data Type	Constraints	Description
1	Admin_id	Varchar(50)	Primary Key	Contains Unique Id
2	Email	Varchar(50)	–	Contains Email Id
3	Password	Varchar(50)	–	Contains Password
4	Name	Varchar(50)	–	Contains Name

Donor

SNO	Column Name	Data Type	Constraints	Description
1	Donor_id	Varchar(50)	Primary Key	Contains Unique Id
2	Donation Details	Varchar(50)	–	Contains Unique Id
3	Password	Varchar(50)	–	Contains Donation Details
4	Name	Varchar(50)	–	Contains Name

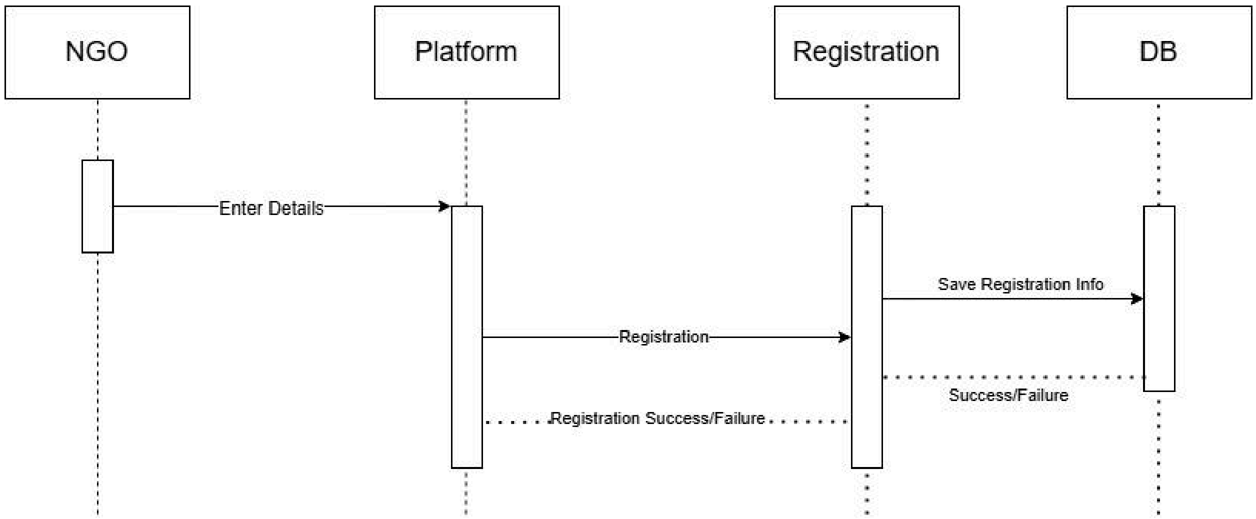
NGO

SNO	Column Name	Data Type	Constraints	Description
1	NGO_id	Varchar(50)	Primary Key	Contains Unique Id
2	Task Details	Varchar(200)	–	Contains Task Information
3	Donation receival	Varchar(200)	–	Contains Donation Details
4	Name	Varchar(50)	–	Contains Name

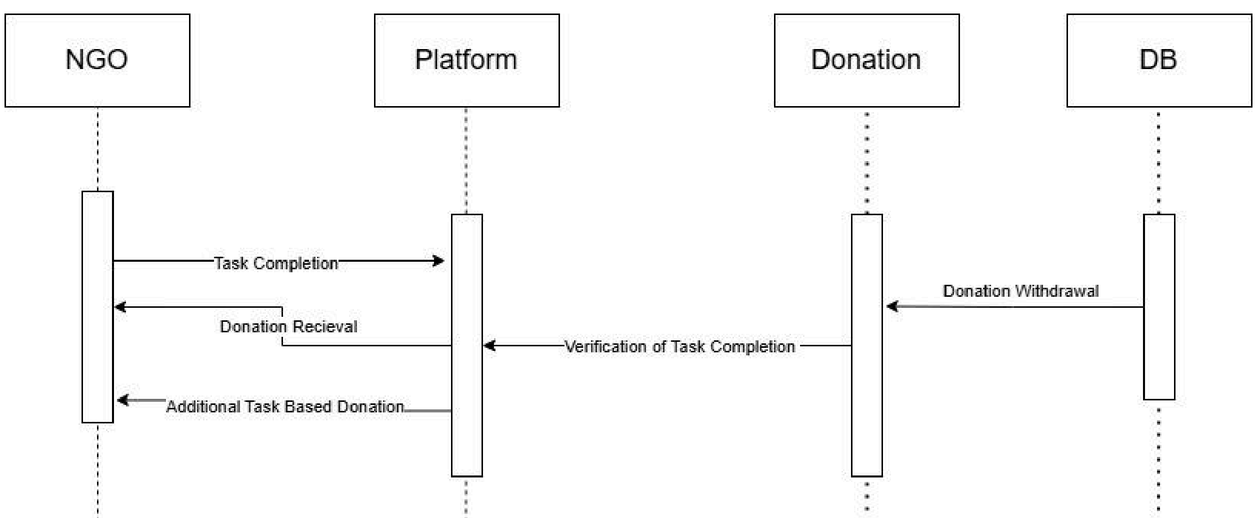
Sequence diagram

NGO MODULE

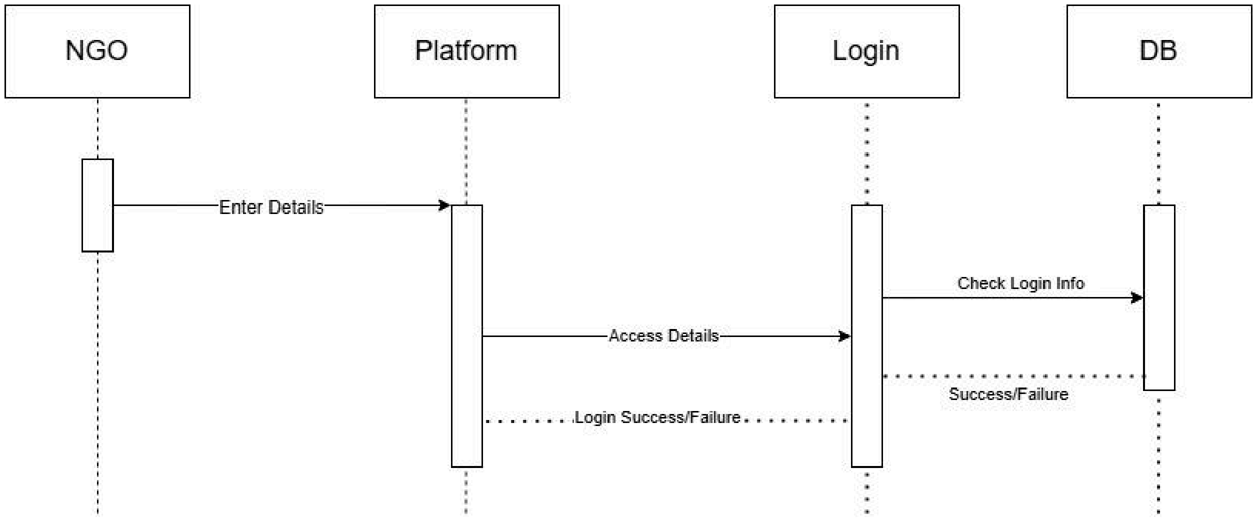
NGO REGISTRATION



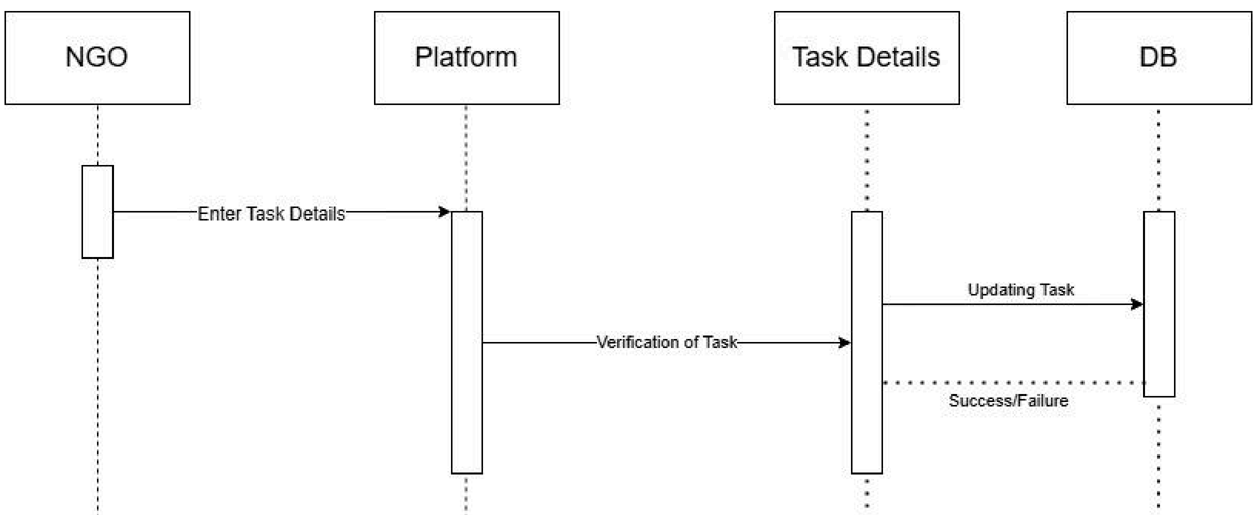
NGO DONATION



NGO LOGIN

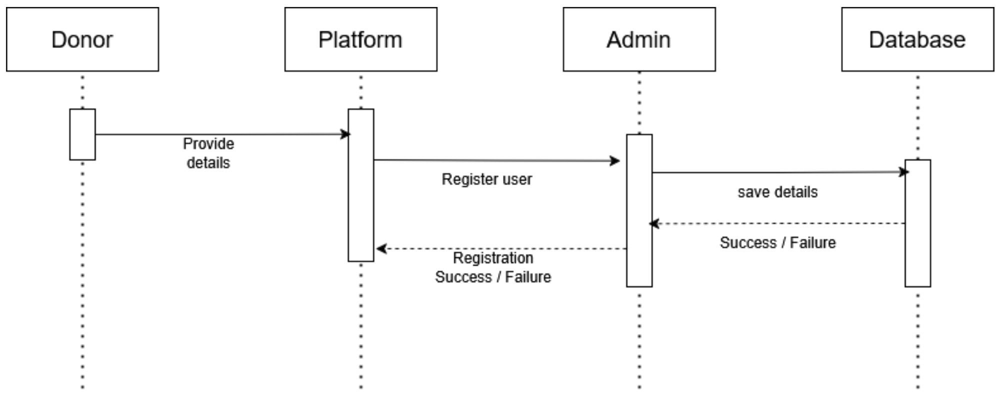


NGO TASK DETAILS

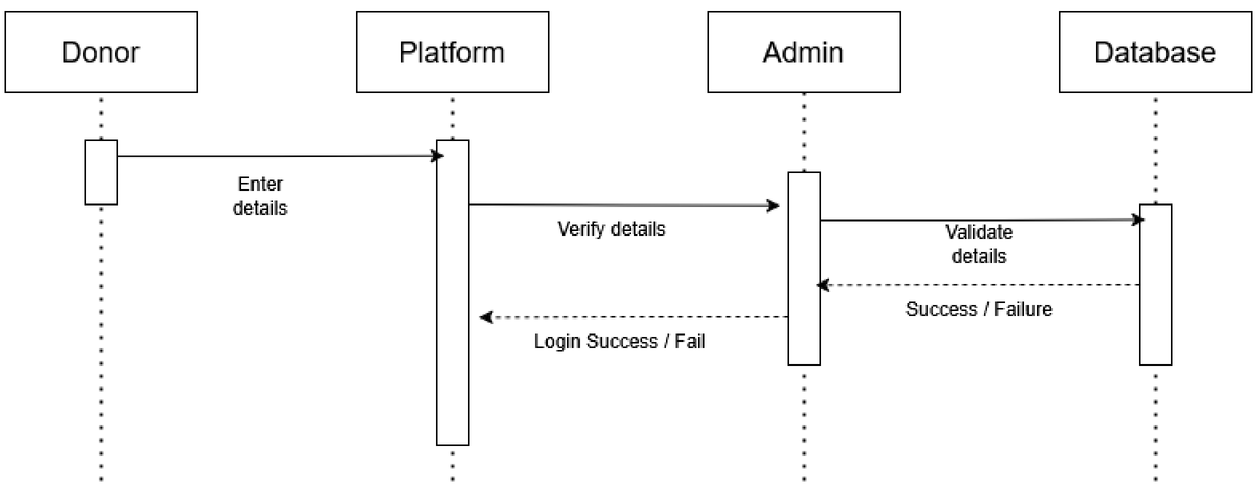


DONOR MODULE

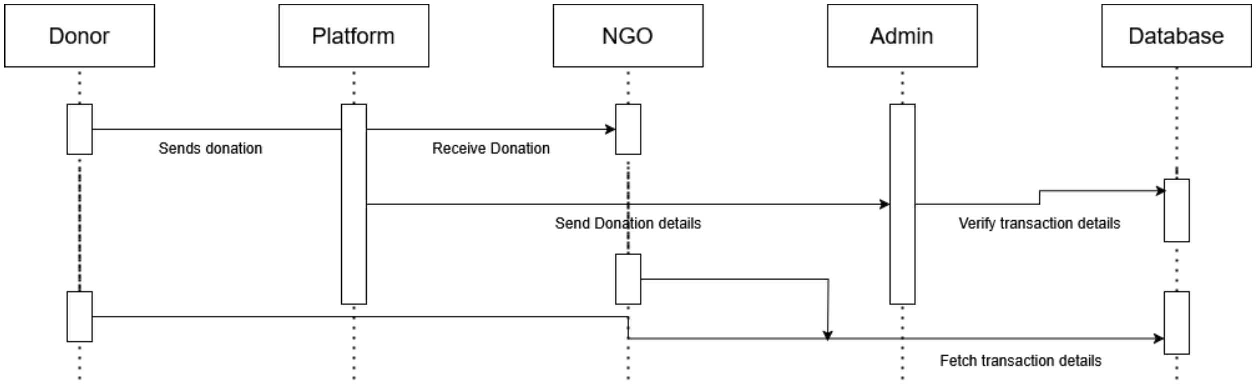
DONOR REGISTRATION



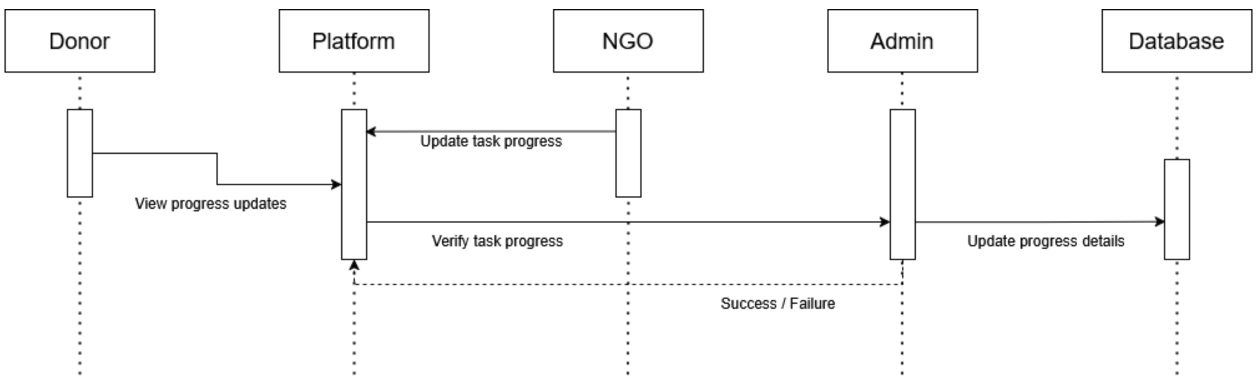
DONOR LOGIN



DONOR DONATION

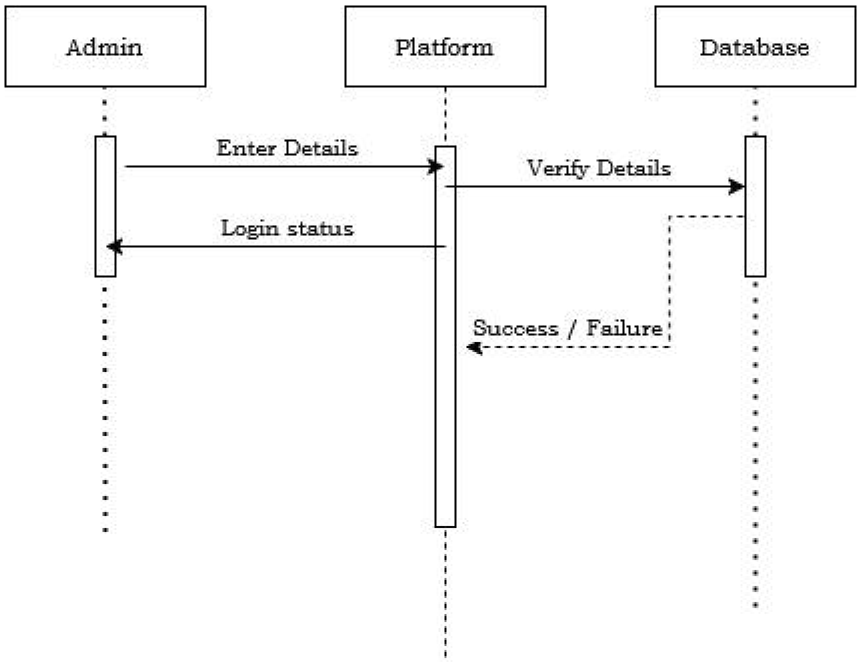


DONOR TASK DETAILS

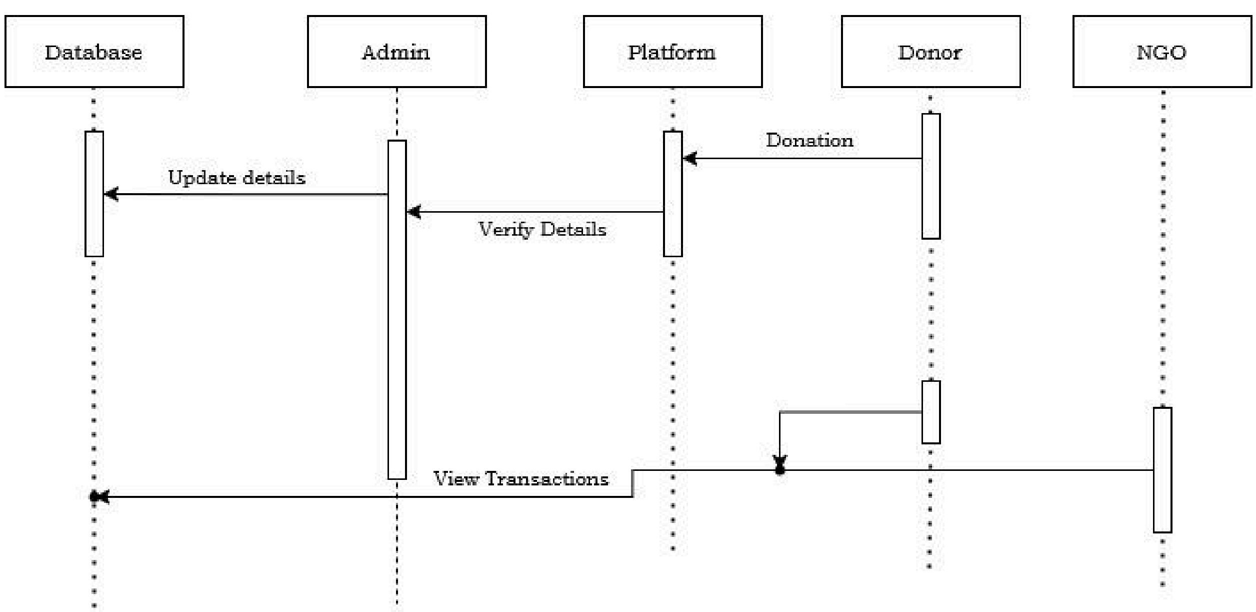


ADMIN MODULE

ADMIN LOGIN



ADMIN DONATION



ADMIN TASK DETAILS